CHAPTER 1

INTRODUCTION

1.1 General Introduction

The rapid growth of the automobile industry has led to the availability of a wide variety of motorcycles with different specifications, performance levels, and price ranges. Choosing the right motorcycle has become a challenging task for customers due to the large number of options available in the market. Most users rely on multiple websites, reviews, and videos to compare motorcycles, making the selection process time-consuming and confusing.

The TrueBiker Researcher is a full-stack web application developed using the MERN (MongoDB, Express.js, React.js, and Node.js) stack to simplify the motorcycle selection process. The application acts as a smart motorcycle recommendation platform that helps users discover motorcycles based on their personal preferences such as riding type, budget, mileage, brand preference, reliability, comfort, and additional features.

The application provides a secure user authentication system using JWT (JSON Web Token) and bcrypt for password encryption. After successful login, users can specify their motorcycle preferences through an interactive interface. Based on these preferences, the recommendation engine analyzes the motorcycle database and suggests the most suitable motorcycles with detailed specifications, match percentage, and recommendation reasons.

The system also provides additional features such as advanced search and filtering, motorcycle comparison, favorite bike management, and a "Know More" option that redirects users to the official manufacturer's website for complete product information. An administrative dashboard is also provided to manage motorcycle records and monitor system activities.

By integrating user preferences with a structured motorcycle database, the application provides personalized recommendations that help users make informed purchasing decisions. The project demonstrates the practical implementation of modern web technologies, database management, authentication mechanisms, and recommendation algorithms in solving a real-world problem.

1.2 Objectives of the Project

The main objectives of the TrueBiker Researcher project are:

- To develop a secure web application using the MERN stack.
- To provide a personalized motorcycle recommendation system based on user preferences.
- To recommend motorcycles by considering factors such as budget, riding style, mileage, reliability, comfort, and brand preference.
- To allow users to compare multiple motorcycles before making a decision.
- To provide detailed motorcycle specifications and feature information.
- To integrate official manufacturer website links through the "Know More" feature.
- To enable users to save their favorite motorcycles for future reference.
- To provide advanced search and filtering options for easy motorcycle exploration.
- To implement secure user authentication using JWT and bcrypt.
- To create an admin dashboard for efficient management of motorcycle data.

1.3 Existing System

Currently, users search for motorcycles using multiple online platforms such as manufacturer websites, review portals, YouTube videos, and comparison websites. These platforms generally provide only product specifications or reviews without offering personalized recommendations based on individual user requirements.

The existing systems have several limitations:

- Users need to visit multiple websites to collect information.
- Recommendation is mostly based on advertisements or popularity.
- Limited personalization according to user preferences.
- Time-consuming comparison process.
- Difficulty in selecting the most suitable motorcycle within a budget.
- Lack of centralized information for different motorcycle brands.
- No intelligent recommendation based on multiple user-selected criteria.

These limitations make the motorcycle selection process difficult, especially for first-time buyers.

Limitations of the Existing System

- Users need to visit multiple websites to collect complete motorcycle information.
- Existing platforms do not provide personalized recommendations based on user preferences.
- Comparing motorcycles from different brands is time-consuming and requires manual effort.
- Recommendations are often based on popularity or advertisements instead of user requirements.
- Limited search and filtering options make it difficult to find suitable motorcycles.
- There is no centralized platform that combines comparison, recommendations, and detailed specifications.
- Most systems do not allow users to save favorite motorcycles or maintain recommendation history.
- Selecting the best motorcycle within a specific budget and riding requirement is difficult for users.

1.4 Proposed System

The proposed system, TrueBiker Researcher, provides an intelligent solution for motorcycle recommendation by analyzing user preferences and matching them with motorcycles stored in the database.

The application allows users to register and log in securely before selecting their preferences, including riding type, budget range, mileage expectations, preferred brands, reliability, and other desired features. Based on these inputs, the recommendation engine calculates a match score and displays the most suitable motorcycles.

Each recommended motorcycle includes complete details such as price, mileage, engine capacity, top speed, features, pros, cons, and match percentage. Users can compare multiple motorcycles, save favorites, and access the official manufacturer website using the "Know More" feature.

The proposed system provides a centralized platform that simplifies motorcycle research and helps users make better purchasing decisions.

Features of the Proposed System

- Secure user registration and login using JWT authentication and bcrypt password encryption.
- Personalized motorcycle recommendations based on user preferences such as budget, riding style, mileage, and brand.
- Advanced search and filtering options to easily find suitable motorcycles.
- Side-by-side motorcycle comparison based on price, specifications, and features.
- Favorites module to save and manage preferred motorcycles for future reference.
- Detailed motorcycle information including engine capacity, mileage, top speed, features, pros, and cons.
- **"Know More"** feature that redirects users to the official manufacturer website for complete product details.
- Admin dashboard to add, update, delete, and manage motorcycle records and monitor application data.

1.5 Feasibility Study

The feasibility study evaluates whether the proposed TrueBiker Researcher system can be successfully developed and implemented. The project is technically feasible because it is developed using the MERN stack, which provides a reliable and scalable platform. It is economically feasible since the required software, frameworks, and development tools are free and open source. The system is operationally feasible because it offers a simple and user-friendly interface for both users and administrators. The project can be completed within the academic semester, making it schedule feasible.

1.5.1 Technical Feasibility

The project is technically feasible because it is developed using modern and widely used technologies such as React.js, Node.js, Express.js, MongoDB, JWT, and Tailwind CSS. These technologies are open-source, well-documented, and suitable for full-stack web application development.

1.5.2 Economic Feasibility

The project is economically feasible because it uses free and open-source software and frameworks. MongoDB Atlas provides a free cloud database, while Vercel and Render offer free deployment plans for educational projects. Therefore, the overall development cost is minimal.

1.5.3 Operational Feasibility

The application is user-friendly and easy to operate. Users can register, log in, select preferences, receive recommendations, compare motorcycles, and explore official motorcycle information through a simple and responsive interface. The system requires minimal user training.

1.5.4 Schedule Feasibility

The project can be completed within the academic semester by following a structured development process, including planning, database design, implementation, testing, and documentation. The modular architecture also allows future enhancements without affecting the existing functionality.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Review – Summary

The increasing popularity of motorcycles has resulted in a significant rise in the number of models available from different manufacturers. Customers often face difficulty in selecting a motorcycle that best suits their riding style, budget, comfort, and performance requirements. To overcome this challenge, several online platforms provide motorcycle information, specifications, reviews, and comparison features. However, most of these platforms primarily focus on displaying technical specifications rather than providing personalized recommendations based on individual user preferences.

Manufacturer websites such as Yamaha, Honda, Royal Enfield, KTM, Bajaj, TVS, Suzuki, Kawasaki, Hero, and other leading brands offer complete details about their own motorcycles. These websites include specifications, pricing, features, image galleries, dealer information, and booking options. However, they are limited to their respective brands and do not allow users to compare motorcycles across different manufacturers.

Motorcycle review platforms provide comparison tools and expert reviews for various motorcycles. Although these platforms help users compare specifications and read reviews, they generally require users to manually search and evaluate different models. Most existing systems do not recommend motorcycles by analyzing a user's riding requirements, preferred brands, mileage expectations, or budget constraints.

Recent developments in web technologies have enabled the development of intelligent recommendation systems that improve user experience by providing personalized suggestions. Recommendation systems are widely used in e-commerce, entertainment, and online shopping platforms to recommend products based on user interests and preferences. Applying similar concepts to motorcycle selection can significantly simplify the decision-making process for customers.

The TrueBiker Researcher project addresses these limitations by integrating a secure authentication system, a structured motorcycle database, and a preference-based recommendation engine into a single web application. Users can specify their riding type, budget range, preferred brands, mileage expectations, and additional features. Based on these inputs, the recommendation engine evaluates the available motorcycles and recommends the most suitable options.

Unlike traditional motorcycle information websites, the proposed system provides a centralized platform that combines personalized recommendations, advanced search and filtering, motorcycle comparison, favorite bike management, and direct navigation to the official manufacturer website

through the "Know More" feature. This approach reduces the time required for motorcycle research and assists users in making informed purchasing decisions.

The project is developed using the MERN (MongoDB, Express.js, React.js, and Node.js) stack, which provides a scalable and efficient architecture for modern web applications. MongoDB is used for storing motorcycle information and user data, Express.js and Node.js manage the backend services and APIs, while React.js delivers a responsive and interactive user interface.

Overall, the literature review indicates that while many existing platforms provide motorcycle information and comparison facilities, there is a lack of systems that combine personalized recommendation, secure user management, intelligent filtering, and centralized motorcycle research in a single application. The TrueBiker Researcher project aims to bridge this gap by providing an integrated solution that enhances the motorcycle selection process and improves the overall user experience.

CHAPTER 3

SYSTEM CONFIGURATION

System Configuration defines the hardware and software requirements needed to develop and run the TrueBiker application efficiently. It ensures smooth performance, secure communication, and reliable operation of the frontend, backend, and database components. The project is developed using the MERN stack and can be executed on standard computer systems with moderate hardware resources.

3.1 Hardware Requirements

Component	**Specification**
Processor	Intel Core i3 (7th Generation) or higher
RAM	8 GB (Minimum 4 GB)
Storage	256 GB SSD or 500 GB HDD
Display	1366 × 768 resolution or higher
Internet Connection	Broadband connection for database access and deployment
Input Devices	Keyboard and Mouse

3.2 Software Requirements

Software	**Purpose**
Operating System	Windows 10/11
Frontend	React.js
Styling Framework	Tailwind CSS
Backend	Node.js
Backend Framework	Express.js
Database	MongoDB
Database ODM	Mongoose

Programming Language	JavaScript (ES6+)
Authentication	JSON Web Token (JWT), bcrypt
HTTP Client	Axios
Routing	React Router DOM
Code Editor	Visual Studio Code
API Testing	Postman
Version Control	Git
Package Manager	npm
Web Browser	Google Chrome / Microsoft Edge
Deployment (Planned)	Vercel (Frontend), Render (Backend), MongoDB Atlas (Database)

CHAPTER 4

MODULE DESCRIPTION

4.1 Modules

The TrueBiker Researcher application is divided into several functional modules. Each module is responsible for performing a specific task, making the system modular, scalable, and easy to maintain. The following are the major modules of the application.

4.1.1 Authentication Module

The User Authentication Module provides secure access to the application by allowing users to register and log in using their email and password.

During registration, user details such as name, email address, and password are collected. Passwords are encrypted using the bcrypt hashing algorithm before being stored in the MongoDB database. During login, JWT (JSON Web Token) authentication is used to verify user credentials and maintain secure user sessions.

Functions

- User Registration
- User Login
- Password Encryption
- JWT Authentication
- Protected Routes
- Session Management

4.1.2 User Preference Selection Module

After successful login, users are redirected to the preference selection page where they can specify their motorcycle requirements.

The system collects user preferences including:

- Riding Type
- Budget Range
- Preferred Brands

- Mileage Requirement
- Reliability Preference
- Comfort Requirement
- Additional Features

These preferences are stored in the database and passed to the recommendation engine.

4.1.3 Motorcycle Database Management Module

This module manages all motorcycle information stored in MongoDB.

The database contains motorcycle details such as:

- Bike Name
- Brand
- Engine Capacity
- Price
- Mileage
- Top Speed
- Fuel Type
- Transmission
- Features
- Pros and Cons
- Reliability Score
- Official Website
- Image URL

The module allows efficient retrieval and management of motorcycle information.

4.1.4 Recommendation Engine Module

The Recommendation Engine is the core component of the application.

It analyzes user preferences and compares them with the motorcycle database to identify the most suitable motorcycles.

The recommendation process considers multiple factors including:

- Riding Style
- Budget
- Mileage
- Brand Preference
- Reliability
- Features
- Beginner Friendliness
- Resale Value

Each motorcycle is assigned a match score, and the highest-scoring motorcycles are displayed to the user.

4.1.5 Search and Filter Module

The Search and Filter Module enables users to explore motorcycles according to specific requirements.

Users can filter motorcycles based on:

- Brand
- Budget
- Mileage
- Engine Capacity
- Riding Type
- Fuel Type
- Reliability

The module provides quick and efficient access to relevant motorcycles.

4.1.6 Bike Comparison Module

This module allows users to compare multiple motorcycles side by side.

The comparison includes:

- Price
- Engine Capacity
- Mileage
- Top Speed
- Features
- Reliability
- Weight

The comparison helps users evaluate different motorcycles before making a purchasing decision.

4.1.7 Favorites Module

Users can save motorcycles that they are interested in for future reference.

The favorite motorcycles are linked to the user's account and can be accessed anytime after login.

Functions

- Save Favorite Bikes
- Remove Favorites
- View Saved Motorcycles

4.1.8 Bike Details and "Know More" Module

Each recommended motorcycle includes a detailed information page displaying complete specifications.

The module also provides a "Know More" button that redirects users to the official website of the motorcycle manufacturer. This allows users to explore complete specifications, available variants, dealer information, and other official details directly from the manufacturer's website.

Functions

- Display Detailed Specifications
- Show Features, Pros and Cons
- Redirect to Official Manufacturer Website

4.1.9 Admin Dashboard Module

The Admin Dashboard provides administrative control over the application.

The administrator can manage motorcycle records and monitor system usage.

The dashboard allows:

- Adding New Motorcycles
- Updating Motorcycle Details
- Deleting Motorcycle Records
- Managing User Information
- Monitoring Recommendation Statistics

This module ensures that the motorcycle database remains organized and up to date.

Functions

- Manage Motorcycle Database
- Update Specifications
- Remove Incorrect Records
- Monitor Users
- View Application Statistics

CHAPTER 5

SYSTEM DESIGN

5.1 Data Flow Diagram (DFD)

A Data Flow Diagram shows how data moves through the system, how it is processed, and where it is stored. It helps in understanding the flow of information between users, modules, and databases.

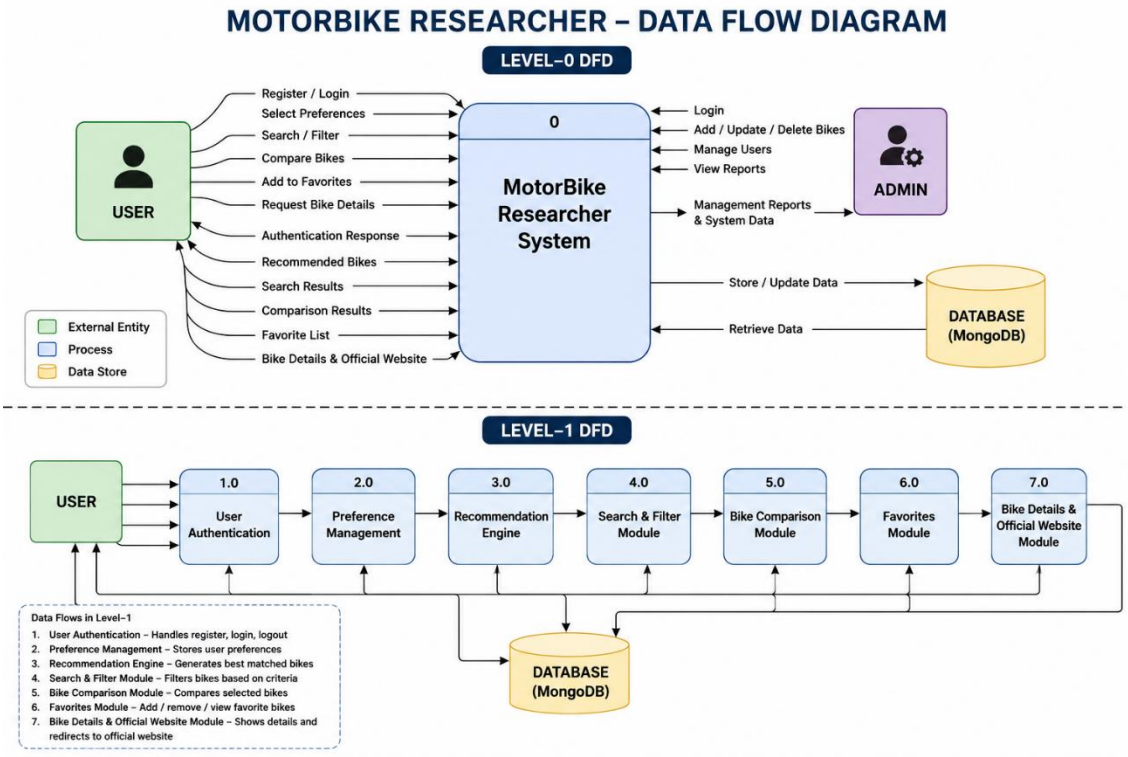


Figure 5.1.1: DFD Diagram

5.2 Use Case Diagram

A Use Case Diagram is a behavioral diagram in the Unified Modeling Language (UML) that shows the interaction between users (actors) and the system. It represents the functional requirements of the system by describing the services provided to different users. The Use Case Diagram helps in understanding the overall functionality of the application from the user's perspective.

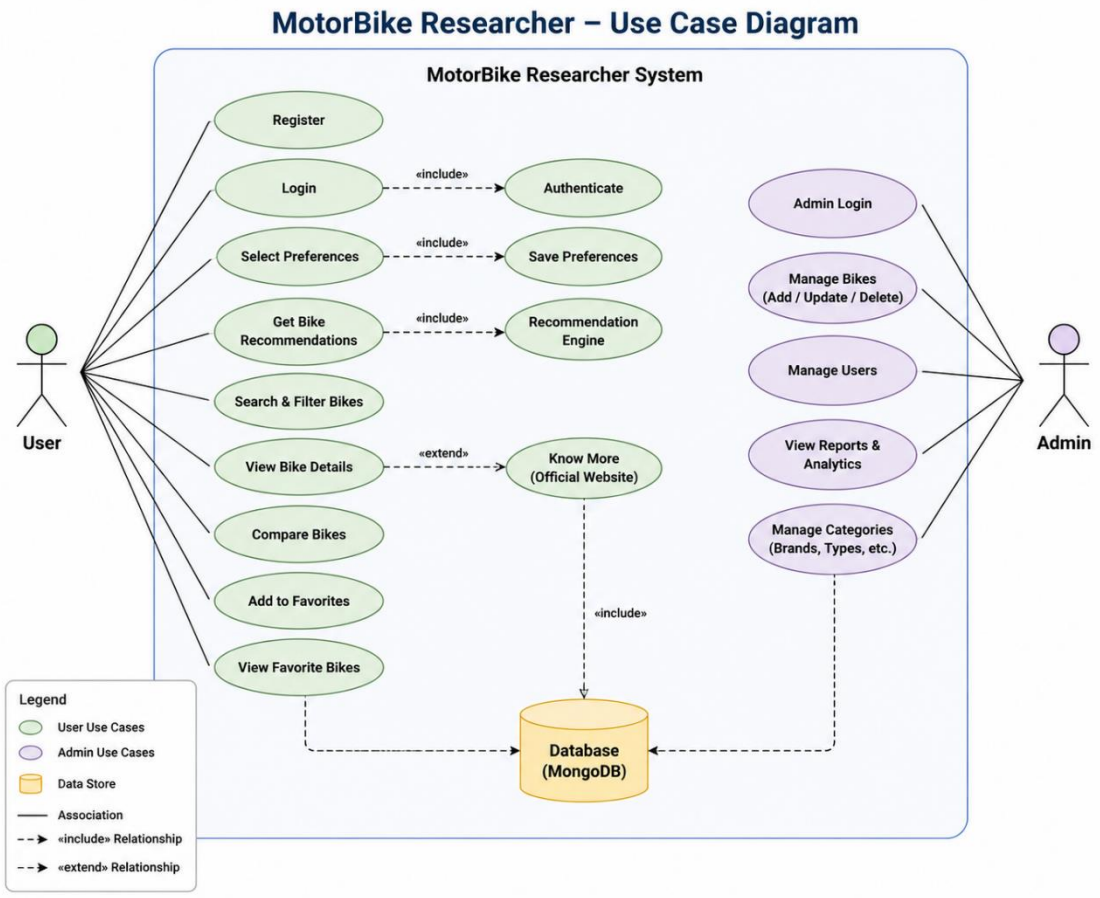


Figure 5.2.1: Use Case Diagram of TrueBiker Researcher

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Implementation

6.1.1 Introduction

System implementation is the phase in which the designed system is transformed into a fully functional application. The TrueBiker Researcher project has been developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The application follows a modular architecture, allowing each component to perform a specific function while maintaining seamless communication between the frontend, backend, and database.

The implementation focuses on providing a secure, responsive, and user-friendly platform for recommending motorcycles based on user preferences**.**

6.1.2 Frontend Implementation

The frontend of the application is developed using React.js and Tailwind CSS. React.js provides reusable components and efficient state management, while Tailwind CSS ensures a responsive and modern user interface.

The frontend consists of the following pages:

- Home Page
- User Registration Page
- Login Page
- Preference Selection Page
- Recommendation Results Page
- Bike Details Page
- Bike Comparison Page
- Favorites Page
- Admin Dashboard

Navigation between pages is handled using React Router DOM. Axios is used to communicate with backend REST APIs.

6.1.3 Backend Implementation

The backend is developed using Node.js and Express.js.

The backend is responsible for:

- User authentication
- API request handling
- Database communication
- Recommendation processing
- Favorite bike management
- Admin operations

Restful APIs are implemented to exchange data between the frontend and MongoDB.

Major API endpoints include:

- User Registration
- User Login
- Get Bikes
- Get Bike Details
- Bike Recommendations
- Save Favorites
- Manage Bikes

6.1.4 Database Implementation

MongoDB is used as the primary database.

The database stores:

User Collection

- User ID
- Name
- Email

- Password (Encrypted)
- Preferences
- Favorite Bikes
- Recommendation History

Bike Collection

- Bike Name
- Brand
- Price
- Engine Capacity
- Mileage
- Top Speed
- Fuel Type
- Transmission

Features

- Categories
- Reliability Score
- Pros
- Cons
- Image URL
- Official Website
- Availability

MongoDB collections are managed using Mongoose schemas for structured data handling and validation.

6.1.5 User Authentication Implementation

The application provides secure authentication using JWT (JSON Web Token) and bcrypt.

Registration

The user enters:

- Name
- Email
- Password
- Confirm Password

Passwords are encrypted using bcrypt before being stored in MongoDB.

Login

During login:

- User credentials are verified.
- JWT token is generated.
- Token is stored securely.
- Protected routes can only be accessed after successful authentication.

This implementation ensures secure user access and protects sensitive information.

6.2 Screenshots

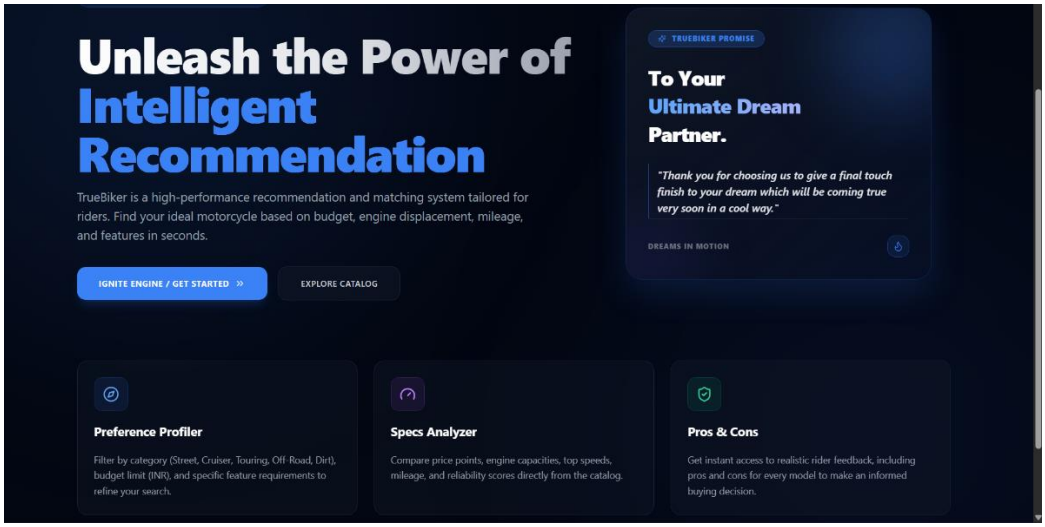


Figure 6.2.1: Home page

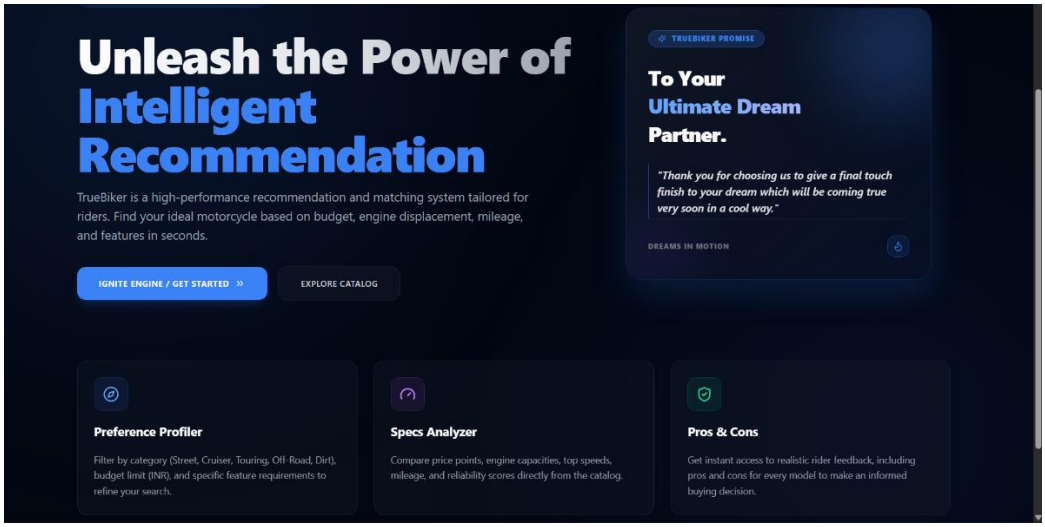


Figure 6.2.2: Sign up

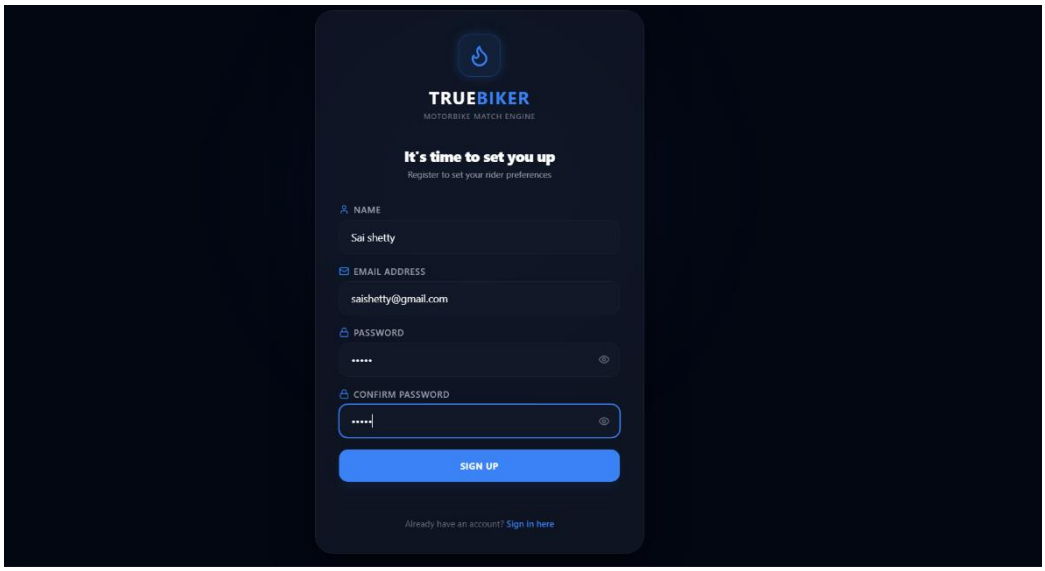


Figure 6.2.3: Login

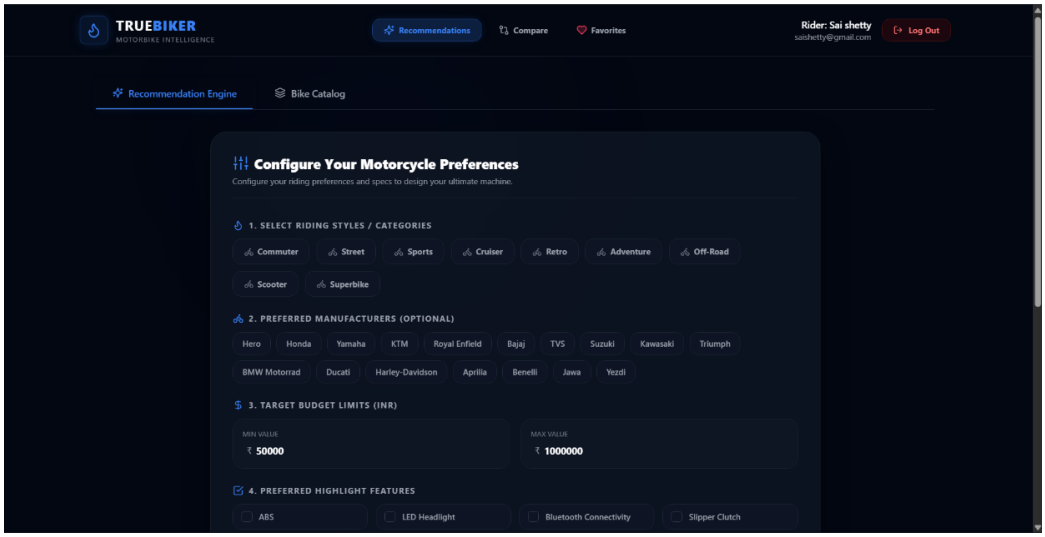


Figure 6.2.4: Preference Selection

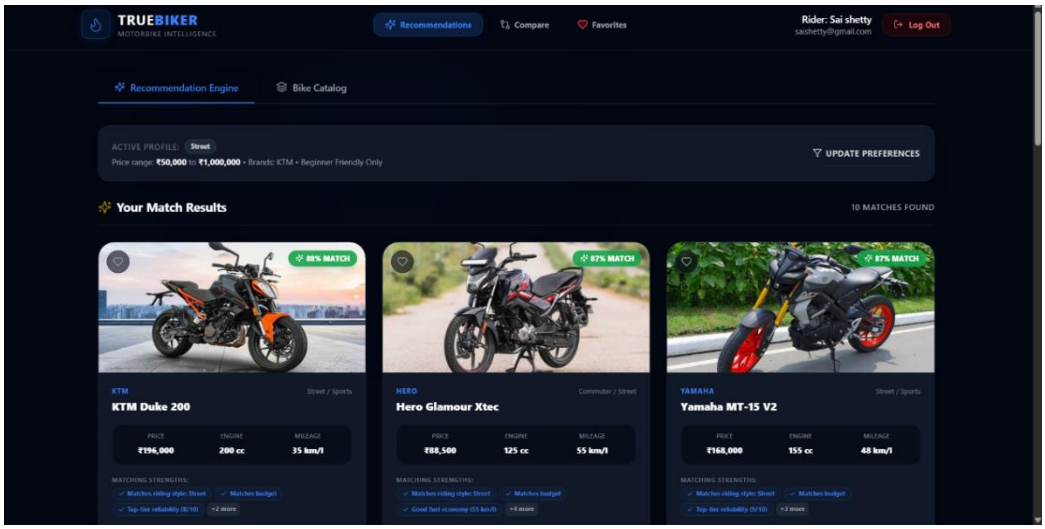


Figure 6.2.5: Recommendation results

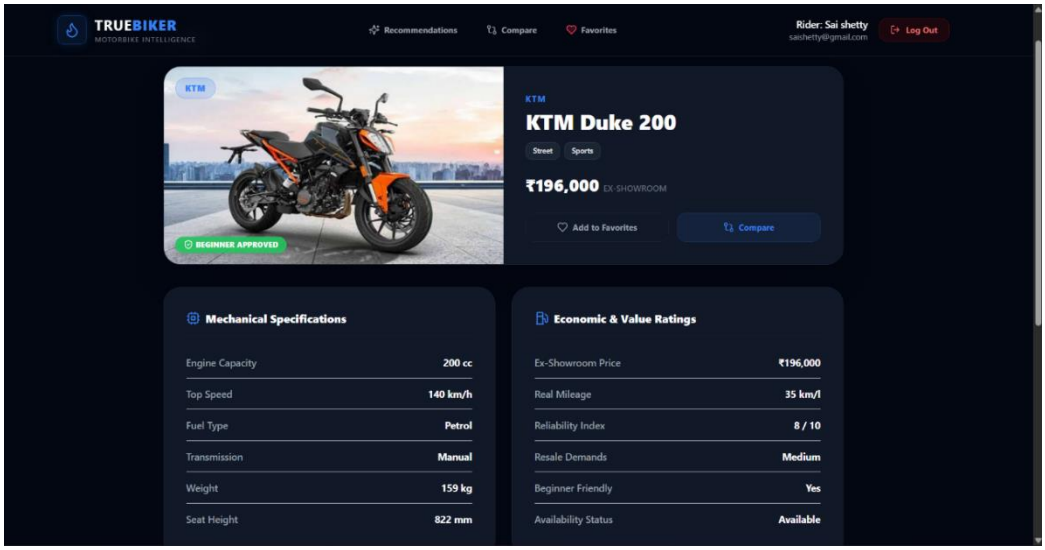


Figure 6.2.6: Bike Details

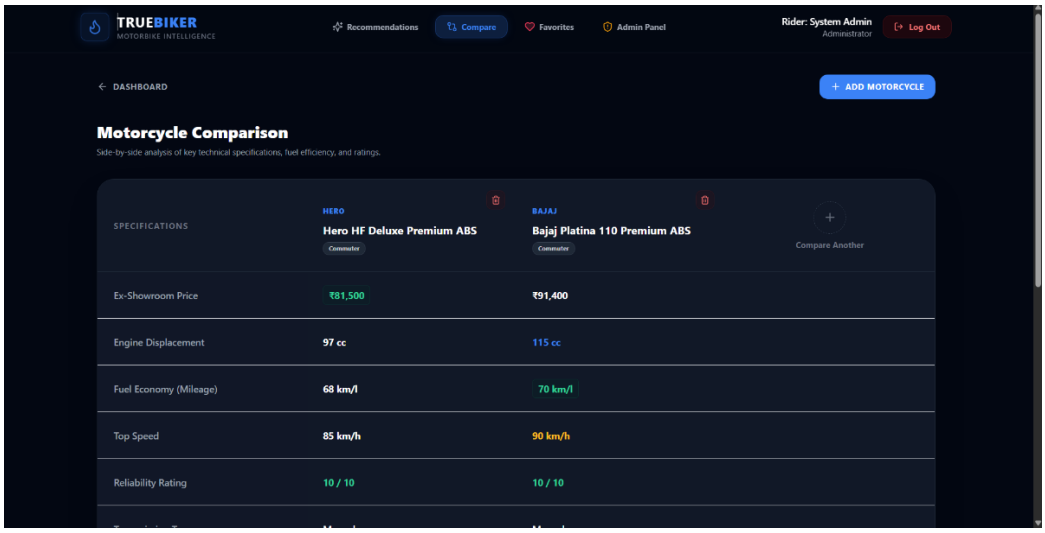


Figure 6.2.7: Bike Comparison

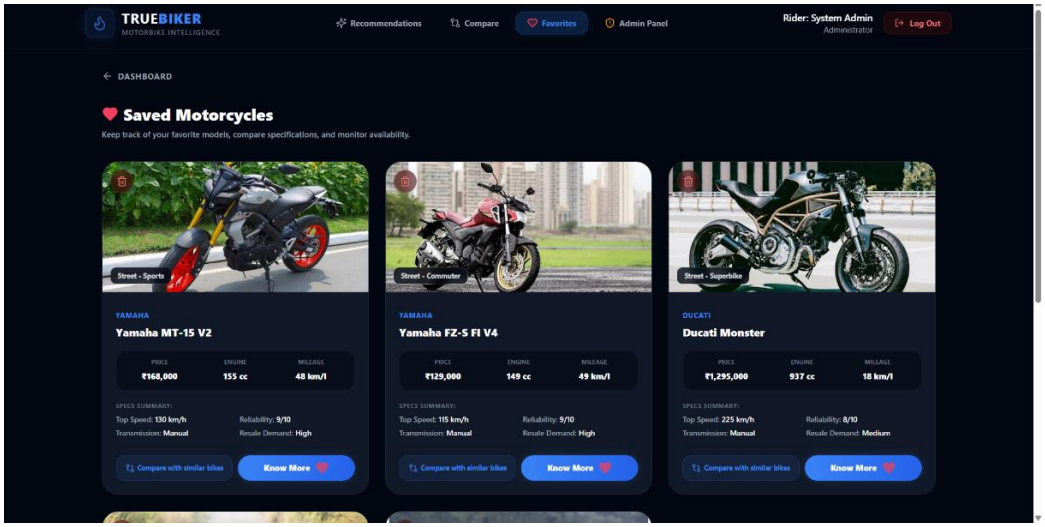


Figure 6.2.8: Favorites

CHAPTER 7

SYSTEM TESTING

7.1 Test Cases

TEST CASES – MOTORBIKE RESEARCHER

Test Case ID	Module	Test Scenario	Test Data / Input	Expected Result	Actual Result	Status
TC-01	Registration	Register with valid details	Name: John, Email: john@gmail.com Password: 123456	User account created successfully	As Expected	Pass
TC-02	Registration	Register with existing email	Email: john@gmail.com (already exists)	Error message: Email already exists	As Expected	Pass
TC-03	Login	Login with valid credentials	Email: john@gmail.com Password: 123456	User logged in successfully	As Expected	Pass
TC-04	Login	Login with incorrect password	Email: john@gmail.com Password: wrong123	Error message: Invalid credentials	As Expected	Pass
TC-05	Authentication	Access protected page without login	Directly access /recommendations	Redirect to login page	As Expected	Pass
TC-06	Preference Selection	Select motorcycle preferences	Riding Type: City, Budget: 1-2 Lakh, Brands: Yamaha, Mileage: 40-50 kmpl	Preferences saved successfully	As Expected	Pass
TC-07	Recommendation Engine	Generate recommendations	Preferences stored in database	Matching motorcycles displayed	As Expected	Pass
TC-08	Search	Search motorcycle by name	Search: "Yamaha R15"	Relevant search results displayed	As Expected	Pass
TC-09	Filter	Apply budget and brand filters	Budget: 1-2 Lakh, Brand: KTM	Filtered motorcycles displayed	As Expected	Pass
TC-10	Compare Bikes	Compare two motorcycles	Select: KTM Duke 390 and Yamaha MT-15	Comparison displayed correctly	As Expected	Pass
TC-11	Favorites	Add motorcycle to favorites	Click on "Add to Favorites"	Motorcycle added to favorites	As Expected	Pass
TC-12	Favorites	Remove motorcycle from favorites	Click on "Remove from Favorites"	Motorcycle removed from favorites	As Expected	Pass
TC-13	Bike Details	View motorcycle details	Select any motorcycle	Complete details displayed	As Expected	Pass
TC-14	Know More	Click "Know More" button	Click "Know More"	Official website opens in new tab	As Expected	Pass
TC-15	Admin Dashboard	Add new motorcycle	Enter new bike details and submit	Motorcycle added successfully	As Expected	Pass
TC-16	Admin Dashboard	Update motorcycle details	Modify bike details and update	Motorcycle updated successfully	As Expected	Pass
TC-17	Admin Dashboard	Delete motorcycle	Select bike and click delete	Motorcycle deleted successfully	As Expected	Pass
TC-18	Logout	Logout from application	Click on Logout button	User logged out successfully	As Expected	Pass
TC-19	Login	Login with empty fields	Email: (empty), Password: (empty)	Error message: Please fill all fields	As Expected	Fail
TC-20	Registration	Register with invalid email format	Email: john@.com	Error message: Invalid email format	As Expected	Fail

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Conclusions

The TrueBiker Researcher project successfully demonstrates the development of a full-stack MERN application that assists users in selecting motorcycles based on their personal preferences. The application integrates secure user authentication, an efficient MongoDB database, a preference-based recommendation engine, advanced search and filtering, motorcycle comparison, favorites management, and direct links to official manufacturer websites.

The project provides a centralized platform that simplifies motorcycle research and reduces the effort required to compare different models across multiple brands. By utilizing modern web technologies such as React.js, Node.js, Express.js, MongoDB, JWT, and Tailwind CSS, the application delivers a secure, responsive, and user-friendly experience.

The modular architecture of the system allows future enhancements such as AI-based recommendations, live motorcycle data synchronization, mobile application support, and advanced analytics. Overall, the project successfully achieves its objectives and demonstrates the practical application of modern web development technologies in solving a real-world motorcycle recommendation problem.

8.2 Future Enhancements

Although the current version of the TrueBiker Researcher application fulfills its primary objectives, several enhancements can further improve its functionality and user experience.

The following features may be implemented in future versions:

- Integration with live motorcycle databases to provide updated specifications and pricing.
- Automatic synchronization of newly launched motorcycles using trusted APIs.
- AI-based recommendation algorithms using machine learning techniques for more accurate suggestions.
- User review and rating system for motorcycles.
- Dealer location and nearby showroom finder using maps.
- EMI calculator and finance estimation for motorcycle purchases.
- Price trend analysis and best time-to-buy predictions.
- Real-time notifications about new launches and special offers.

- Voice-based motorcycle search and recommendation.
- Mobile application development for Android and iOS platforms.
- Multi-language support to improve accessibility for users from different regions.
- Advanced analytics dashboard for monitoring user preferences and recommendation statistics.
- Integration with cloud storage and content delivery networks (CDNs) for optimized image management.
- Improved recommendation scoring using historical user interactions and recommendation history.

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